An hourly kilometre-scale fine particulate matter reconstruction for Kandy, Sri Lanka: construction, validation without local ground truth, and measurement priorities

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The problem

Fine particulate matter (PM2.5) carries the largest estimated burden of disease of any air pollutant, and monitoring is sparsest where concentrations are highest. Kandy is such a city: a valley settlement in the central highlands of Sri Lanka with 98,828 residents inside its municipal boundary (2012 census) and nearly 389,000 commuters entering on a typical weekday (World Bank, 2020), so that most of those exposed to its air during the day do not live in it. It has no reference-grade PM2.5 monitor in operation and two low-cost sensors with public records. Any PM2.5 field for Kandy is therefore a model output that almost no local observation can evaluate.

Aim

To reconstruct PM2.5 over the Kandy basin at hourly and one-kilometre resolution for 2019 to 2023 from data available everywhere; to establish how far each part of that reconstruction can be trusted; to determine the spatial scale below which it is no longer supported; and to identify the measurements that would most improve it.

The model

Concentration on the kilometre grid is a regional background, uniform across the city at any hour, plus a local increment redistributed by a pattern of unit spatial mean:

$$\mathrm{PM}(x,y,t) = B(t) + \max(\Delta,0)\,P(x,y,t) + \min(\Delta,0) + \varepsilon(t)\,[P(x,y,t)-1], \qquad \Delta(t) = T(t) - B(t)$$

T, the basin-mean concentration, is a gradient-boosted model of the residual between the two local sensors and a composition-model prior, driven by reanalysis meteorology and satellite retrievals, shifted each year to a satellite-derived area mean and shaped to the sensors' daily and seasonal cycles. B is a regional-background estimate built from the same satellite level, the composition model and an air-mass classification, capped so that it never reaches the hourly total. P combines a satellite surface with a terrain-confinement term; a terrain-steered dispersion step was removed after it lowered agreement with held-out stations. The field averages to T (to within 0.0 per cent in practice), so misplacing material cannot create more of it, and each component declares which observations it may use. The local increment is a residual of this decomposition, not the mass emitted inside the city.

Validation without local ground truth

The two Kandy sensors trained and shaped T, so they cannot validate it. Three other lines of evidence are used, and their results are graded rather than combined. The complete model was built in ten monitored valley and basin cities with only two anchor stations, as at Kandy, and scored against withheld stations: the seasonal cycle transfers everywhere (correlation 0.943 to 0.996), the daily cycle unevenly (-0.325 to 0.977). The deployed temporal model was placed on a cross-city ladder of information budgets in 104 cities (exploratory, specified before scoring): where its anchor stations observed the period it cut daily error by 32.7 per cent against a generic sensorless estimate (correlation 0.864); where they did not, by 4.83 per cent, not distinguishable from zero. Its satellite-derived level sat 7.87 per cent above the withheld city means and its nominal 90 per cent interval covered a fraction 0.696. At Kandy the national research organisation's daily record for 2021 and 2022, recovered from a published figure, matches the field's seasonal shape (correlation 0.77 across calendar months) but not its day-to-day variation (within-month correlation 0.284), and the field misses the crisis year 2022. Trained on one Kandy sensor, the temporal model tracks the other at a daily correlation of 0.521 and 0.615. Against the regulatory station in Kandy city, whose public record covers 858 days of 2019 to 2023, the field's daily correlation is 0.6 (0.72 within the month).

The reconstructed field, and how far it can be trusted

The modelled annual basin mean is 17.0 to 20.9 micrograms per cubic metre over 2019 to 2023, above the World Health Organization guideline of 5 in every year. The level is probably too high: the regulatory station reads 24 per cent below the field, as does a calibrated low-cost record, while the national research record, of undocumented instrument, sits above it. The excess enters where the anchor's level is set from the satellite surface: the corrected low-cost record is within -5 per cent of the station on shared days. The exceedance is robust to it: if the satellite level errs at Kandy as it does in the middle half of the tropical cities on the ladder, the basin mean is 13.5 to 22.1, at least 2.7 times the guideline. The seasonal cycle is the best-supported property, with the maximum in the north-east monsoon (30.2 in December to February of 2023) and the minimum in the south-west monsoon (11.7). The day-to-day sequence is supported on days the sensors reported. Morning and evening peaks either side of a midday dip are robust in timing, but their depth depends on the humidity correction of the sensor record: with hourly rather than constant humidity the morning-peak-to-midday ratio falls from 1.93 to 1.68 and the night falls below midday, while the level, the daily sequence and the local fraction barely move. The regulatory station's hourly record has the same timing and a morning peak like the delivered field's, but its deep night is the daily minimum (0.71 of midday), so the field's nights are too high. The interval covers 72.5 per cent of sensor hours with misses almost all on one side, which points to a calibration bias at the sensors rather than proving the width right. Under the baseline decomposition the local increment is 0.483 of the modelled total, 0.433 to 0.492 across the tested ways of applying the cap and 0.547 with a 48-hour background window: a property of the decomposition, not a source apportionment.

What each additional observation is worth

Because every component declares its inputs, an observation can be withheld and the loss measured across monitored cities. A registered confirmation on 72 fresh cities found two stations used only to recalibrate a free estimate worth 8.53 per cent and a background series read daily worth 41.1 per cent. A later review showed the comparison was not like for like. In a post hoc re-analysis that used every source on the day, one station read daily reduced daily error by 42.1 per cent, two by 53.3 and five by 58.4, against about 11.0 per cent for stations used only as a calibration, and no ordering of station types was established. For Kandy the priority is continuously reporting stations whose readings reach the estimate every day, with one of reference grade to settle the level and the humidity correction.

Limitations

The field is not supported below the kilometre scale. Two sites three hundred metres apart inside one model cell differ in measured PM10 by a factor of 27.5, though on different days, so the pair illustrates sub-grid variation without isolating location; on densely monitored cities the spread inside a cell (1.22) exceeds the spread between cells (1.04). A dispersion step meant to place the local increment ranked held-out stations at 0.274, below the undispersed surface (0.371) and a free built-up layer (0.393), and was removed; three to eight stations do not rank neighbourhoods usefully either. The map is a hypothesis for testing, not a ranking of neighbourhoods. The chemical check is a consistency check within a composition model, not a chemical measurement at Kandy.

Ongoing work and verification

A formal agreement for the regulatory record, and co-locating the low-cost sensors with it, would fix the size of the level excess and show where the sensor record's nights go wrong. A fresh pre-registered like-for-like test and an independent reproduction are the next steps for the cross-city results. Every value in the report is regenerated from source at build time by a gate that refuses to build when prose and data disagree (github.com/daminda1108/kandy_pm2.5_framework); twenty-three of 105 registered predictions were refuted and are reported as such.

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